DIGITAL COURSE PORTFOLIO

IS132-1L • Project Management 1 Laboratory



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Student Number: 2021112008

Section A323 | 1st Semester AY 2026-2027

PROJECT

FLOWBIKES: A Case Study of Project Risk Monitoring
and Schedule Dependency Tracking in a Web-Based Bicycle Sales System

Instructor: Cherry B. Lisondra | Mapúa Malayan Colleges Mindanao

Course Introduction & Learning Goals

Course Overview

Project Management 1 Laboratory guides students through transforming System Analysis and Design outputs into fully executed Information Systems projects. Students gain practical experience in project lifecycles, rotating leadership roles, stakeholder engagement, risk management, and schedule control within real academic constraints.

01

Transform SAD to Baselines

Convert System Analysis and Design outputs into structured IS project artifacts including Project Charters, scope statements, and feasibility matrices.

02

Manage Risks & Dependencies

Apply risk monitoring and schedule dependency tracking to identify bottlenecks, manage technical risks, and support control decisions.

03

Execute Lifecycle Controls

Lead and monitor an Information Systems project from initiation through risk management, cost control, testing, and formal closure.

Project Overview

FLOWBIKES – Web-Based Bicycle Sales System

FlowBikes is a web-based bicycle sales system developed by the SPR DEVS team. The system supports product browsing, specifications viewing, AI-assisted customer inquiries and product suggestions, map information, and weather data. Administrative functions are restricted to an admin account; customers do not require accounts. The system does not process online payments—customers are directed to the business Facebook page for purchase communication.

Key Focus of the Case Study

This project served as a single-case study examining how project risk monitoring and schedule dependency tracking supported control decisions. The team used WBS, Responsibility Assignment Matrix (RAM), Activity Network Diagram, and risk management documents to organize work, assign responsibilities, identify risks (schedule delays, limited resources, technical integration, third-party services, scope changes, AI accuracy), and document monitoring practices.

My Project Manager Rotation

WEEK 5

Introduction to Planning, Performing and Controlling

Role: Student C – Project Manager

Key Responsibilities
- Supported project planning framework
- Helped establish performance baselines
- Contributed to activity sequencing and dependency mapping
- Assisted in preparing monitoring and control structures

WEEK 10

Risk Management and Project Monitoring

Role: Student C – Project Manager

Key Responsibilities
- Led risk identification and documentation
- Supported risk register maintenance
- Monitored schedule dependencies
- Contributed to progress tracking and corrective-action discussions

My Project Contributions

Planning & Baselines (Week 5)

Assisted in mapping the project management plan framework, defining performance baselines, and supporting activity dependency identification for the FlowBikes development lifecycle.

Risk Monitoring (Week 10)

Contributed to the identification and documentation of schedule, resource, technical, and third-party risks. Supported risk register updates and discussions on measurable triggers.

Research Paper Contribution

Served as a contributor to the final research paper examining project risk monitoring and schedule dependency tracking. Helped organize evidence from WBS, RAM, Activity Network Diagram, and risk documents.

Schedule Dependency Awareness

Participated in reviewing task sequences and highlighting how delays in upstream activities (requirements, design, development) could cascade into testing and documentation phases.

Evidence of Work

Key artifacts supporting personal contribution (placeholders for screenshots / file links)

Week 5

IS132-1L – PROJECT MANAGEMENT 1 LABORATORY

ACTIVITY 5 – INTRODUCTION TO PLANNING, PERFORMING AND CONTROLLING

WEEK 5 | Project Management Plan Framework | PM Rotation: Student C

Topics: Planning, performing, monitoring, controlling, project baselines, performance, and change management.

Purpose: Build the framework that will organise all future project-management plans and establish how the project will be performed, monitored, controlled, and changed.

A. PROJECT INFORMATION

Field	Team Response
Course / Section	IS132-1L – Project Management 1 Laboratory
Team Name	MAAFF
Project Title	FLOWBIKES: A Web-Based Bicycle Sales System
Project Manager	Student C
Proposal Used	FLOWBIKES SAD Proposal
Current Status	☑ On Track ☐ At Risk ☐ Delayed

B. PLANNING MAP

Management Area	What Must Be Planned?	Key Decision	Planned Artifact	Owner
Scope	Define system scope, features, and limitations.	Keep core features only.	SAD Scope and Limitations	Project Team
Quality	Define functional, usability, and data-quality standards.	Test core functions before acceptance.	Quality Plan / Test Cases	Project Team
Team / Responsibilities	Assign tasks between researchers and owner.	Clarify responsibilities and deliverables.	Responsibility Matrix	Project Team
Activities / Work	Plan development, testing, documentation, and presentation.	Follow the approved WBS.	WBS / Activity Plan	Project Team
Schedule	Set milestones for development and completion.	Track weekly progress.	Gantt Chart / Schedule	Project Team
Resources	Identify people, software, hardware, APIs, and hosting.	Use available tools efficiently.	Resource Plan	Project Team
Cost / Budget	Estimate project-related expenses and service costs.	Prioritize available/free resources.	Cost Estimate	Project Team

Planning Framework & Baseline Documents

Week 10

IS132-1L – PROJECT MANAGEMENT 1

WEEK 10 – RISK MANAGEMENT AND PROJECT MONITORING

PM Rotation	Student C
Topics	Risk identification; probability; impact; response; risk register; monitoring.
Activity	Identify technical, schedule, resource, cost, team, requirement, and external risks; prepare the risk register and response plan.
Deliverable	Risk Register + Risk Response Plan

A. INSTRUCTIONS

1. Review scope, schedule, resources, and budget for risks.
2. Record probability and impact.
3. Assign a risk owner.
4. Define response and trigger/condition when appropriate.
5. Define monitoring actions.
6. Update the Research Log with the risk issue, decision, result, lesson, and evidence.

B. WORKING DOCUMENT TEMPLATE

1. Risk Register

ID	Risk	Category	Probability	Impact	Response	Owner	Trigger	Monitoring
R 1	Delay in core module development affecting integration	Schedule / Technical	Medium	High	Mitigate: Prioritize core modules; monitor weekly progress	Francis (PM)	Core modules incomplete by planned end date	Weekly status check vs Activity Schedule (Week 8)
R 2	Resource bottleneck: same members on development, integration, and testing	Resource	Medium	Mediu m	Mitigate: Distribute via RAM; coordinate overlapping work	Francis / Team	One member assigned to 3+ concurrent critical tasks	Review RAM and workload at each milestone
R 3	Free-tier limits (Vercel/Supab ase) exceeded during testing or deployment	Technical / Cost	Low	Mediu m	Mitigate: Monitor usage; contingency ₱1,000 reserved	Keith / Dev Team	Free-tier quota warning or rate limit	Check dashboard usage weekly; keep contingency
R 4	AI/API recommendati on service cost exceeds	Cost	Low	Mediu m	Mitigate: Limit test calls; use free alternatives	Development Team	API spend approaches ₱800	Track API usage against Week 9 budget

Risk Register & Monitoring Notes

Research

Project record	Documented issue or decision	Recorded response / implication
Project Charter and Scope Statement	The project established the objectives, system requirements, and boundaries of the FlowBikes web-based bicycle sales system.	The documents provided the foundation for controlling project scope and defining the system's intended features and limitations.
Work Breakdown Structure (WBS)	Project activities were divided into smaller tasks covering the planned development process.	The WBS provided a structure for organizing work, assigning responsibilities, and monitoring planned activities.
Responsibility Assignment Matrix (RAM)	Responsibilities were distributed among the project team members.	The RAM clarified task ownership and provided a basis for coordinating individual contributions.
Activity Network Diagram	Relationships and dependencies between project activities were mapped.	The diagram illustrated the intended task sequence and helped identify activities that could affect subsequent work.
Risk Management Records	Potential schedule, resource, and technical risks were identified, including integration challenges and third-party service dependencies.	Risk identification provided a basis for planning monitoring activities and appropriate response strategies.
Project Monitoring Records	Available consolidated records did not provide sufficient dated progress updates or complete evidence of implemented corrective actions.	Actual schedule performance and the effectiveness of risk responses could not be conclusively established.
Project Research Records	The available documents were reviewed to examine project planning, risk management, and schedule dependency tracking.	The analysis identified documentation gaps and provided recommendations for improving future project monitoring and control.

Table 2. Summary of the documented project planning, risk management, and monitoring records for FlowBikes.

Discussion

The FlowBikes project planning documents address different areas that may affect project delivery. The WBS provides a structured breakdown of activities to help organize and monitor project work. The RAM establishes responsibility assignments to clarify task ownership and improve coordination among team members. The Activity Network Diagram illustrates dependencies between activities, helping the team identify tasks whose delays could affect subsequent work. Risk management records also provide a basis for identifying potential schedule, resource, and technical problems before they affect project objectives.

These practices are consistent with software project risk management principles, which emphasize identifying and addressing potential risks throughout the development process (Boehm, 1991; Keil et al., 1998). The use of activity dependencies and coordinated task planning also supports schedule management by helping project teams understand how

Final Research Paper (Contributor)

Project Management Learning

01

Risk Identification & Monitoring

Learned that risks involving schedule delays, limited resources, technical integration, third-party services, and AI accuracy must be documented early and paired with measurable triggers rather than left as static lists.

02

Schedule Dependency Tracking

Understood that the Activity Network Diagram and WBS are only effective when supported by actual progress records. Dependencies become actionable only when start/finish dates and status updates are maintained.

03

Evidence Gaps in Control

Recognized that planning documents alone cannot prove performance. Dated progress logs, revised effort figures, and verified testing results are required to evaluate whether risk responses actually worked.

04

Documentation Discipline

Appreciated the value of consolidating evidence trails (WBS, RAM, risk registers, decision logs) so that future teams can distinguish planned actions from verified outcomes.

Research Contribution

Role: Research Paper Contributor

I contributed to the single-case study research paper titled “FLOWBIKES: A Case Study of Project Risk Monitoring and Schedule Dependency Tracking in a Web-Based Bicycle Sales System.”

Specific Contributions

- Supported the consolidation of project management artifacts (Project Charter, WBS, RAM, Activity Network Diagram, risk documents) into an evidence matrix.
- Helped identify and categorize documented risks related to schedule, resources, technical integration, third-party services, scope changes, and AI-generated content accuracy.
- Assisted in distinguishing planned management practices from verified outcomes, highlighting documentation gaps that limited performance evaluation.
- Contributed to the discussion of recommendations for future student projects: regular progress records, updated risk registers, dependency monitoring, and documented corrective actions.

Individual PM Reflection

Challenges	Maintaining clear visibility of task dependencies and ensuring risk items were paired with measurable triggers rather than remaining as general concerns.
Decisions	Supported the prioritization of core development and testing activities while keeping out-of-scope expansions under control. Emphasized the need for dated progress updates to enable real monitoring.
Communication	Participated in structured discussions that aligned technical work with schedule and risk status so that the team shared a common understanding of critical path and potential bottlenecks.
Teamwork	Worked with teammates to consolidate scattered documents into a usable evidence base for the research paper, ensuring individual contributions remained visible within the group effort.
Lessons Learned	Planning documents are necessary but insufficient. Continuous progress logging, risk-register updates, and explicit documentation of corrective actions are what turn monitoring into effective control.

Course Achievements

Achievement / Certificate	Provider	Date	Evidence
Project Management: Foundations and Initiation	University of Colorado Boulder (Coursera)	September 18, 2026	CERTIFICATE LINK


University of Colorado Boulder

COURSE CERTIFICATE

Sep 18, 2026

Josh Marcus Amahan

has successfully completed

Project Management: Foundations and Initiation

an online course authorized by University of Colorado Boulder and offered through Coursera

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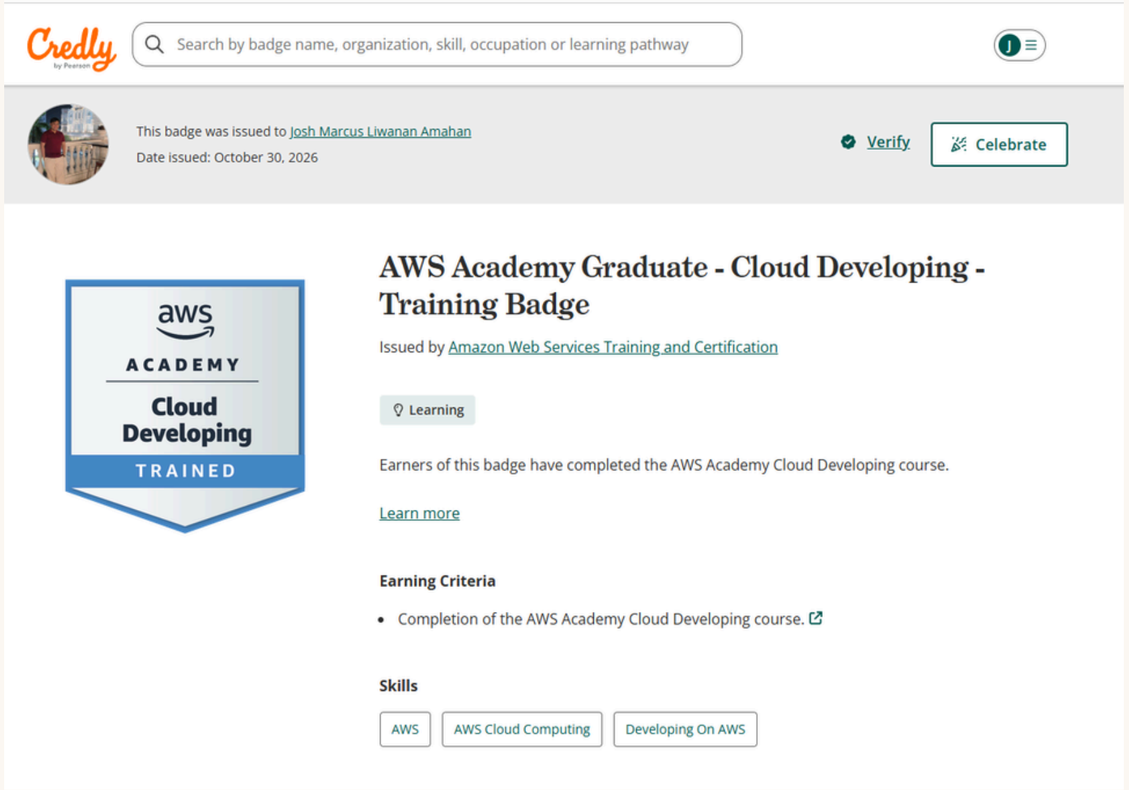
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Lockheed Martin Professor of Engineering Management
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Course Achievements

Achievement / Certificate	Provider	Date	Evidence
AWS Academy Graduate – Cloud Developing Training Badge	Amazon Web Services	October 31, 2026	BADGE LINK



AI Use and Verification

AI Tool	Purpose	Output Used	Verification	Status
ChatGPT	Drafting outlines, reflection structure, evidence organization	Initial structure for reflection and contribution statements	Cross-checked against project records and team decisions	Verified
Gemini	Fact-checking risk categories, schedule dependency language, literature alignment	Refined risk descriptions and recommendation wording	Compared with WBS, RAM, and research paper evidence	Verified

All AI-assisted content was reviewed, revised, and validated against actual project artifacts. Final decisions and conclusions remain the student's responsibility.

Final Reflection

Managing an Information Systems project reinforced that effective control depends on continuous evidence rather than initial plans alone. Through the FlowBikes case study I learned that risk registers and schedule dependency diagrams become useful only when they are kept current with dated progress updates, revised effort figures, and documented corrective actions.

Serving as Project Manager during the planning/control and risk-monitoring weeks highlighted the practical difficulty of distinguishing planned responses from verified outcomes. The research contribution further emphasized the importance of transparent documentation so that teams and evaluators can assess what was actually monitored and adjusted.

Moving forward, I will prioritize early establishment of progress-logging habits, measurable risk triggers, and clear ownership of dependency-critical tasks. These practices strengthen both academic project delivery and professional readiness for real-world Information Systems work.

Evidence Index / References

Primary Project Artifacts

- Project Charter, Scope Statement, Work Breakdown Structure (WBS)
- Responsibility Assignment Matrix (RAM)
- Activity Network Diagram
- Risk management documents and risk register entries
- Final Research Paper – FLOWBIKES case study (Contributor)
- Week 5 Planning / Performing / Controlling materials
- Week 10 Risk Management and Project Monitoring materials

Selected Literature References

Boehm, B. W. (1991). Software risk management: Principles and practices. IEEE Software.
Keil, M., et al. (1998). A framework for identifying software project risks. Communications of the ACM.
Fleming, Q. W., & Koppelman, J. M. (2010). Earned value project management (4th ed.). PMI.

E. Individual Portfolio Evidence Table

Week	Role / Contribution	Artifact / Evidence	What I Learned	File Link
5	PM (Student C) – Planning & Control	Planning framework, baselines	Baselines enable later variance tracking	WEEK 5 – INTRODUCTION TO PLANNING, PERFORMING AND CONTROLLING
10	PM (Student C) – Risk & Monitoring	Risk register, dependency notes	Triggers + ownership make risks actionable	WEEK 10 – RISK MANAGEMENT AND PROJECT MONITORING
12-14	Research Contributor	Project Management Research Review, Research Study & Presentation Preparation, and Final Research Paper	Evidence gaps limit performance claims	MAAFF_IS132-1L_W12_Research_Proposal_Initial_Literature_Review MAAFF_IS132-1L_Week13_Research_Study_Preparation IS132-1_ResearchPaper_MAAFF